DBMS

PROJECT

- BY-

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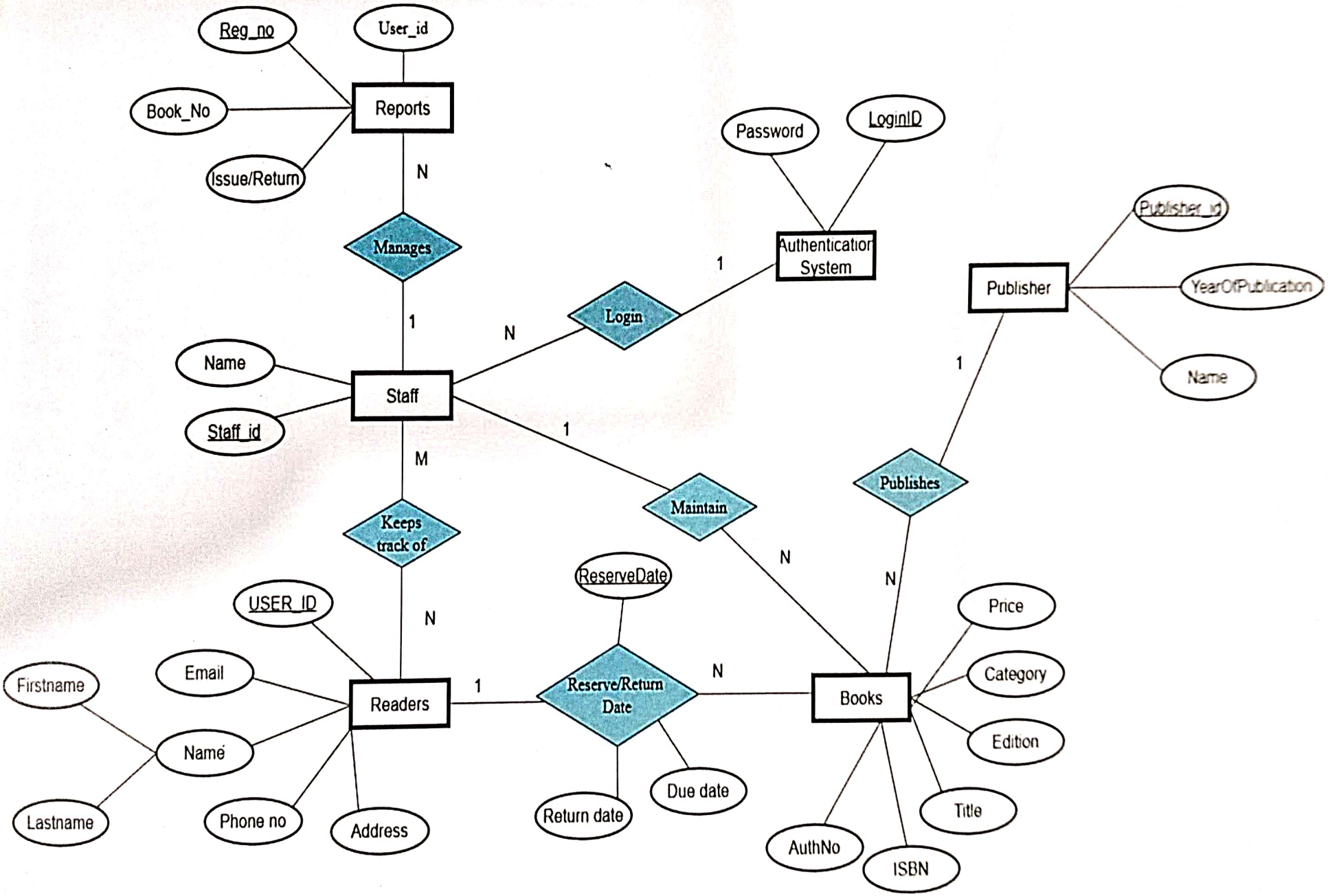
- CLASS → 2CSE24
- TOPIC → LIBRARY MANAGEMENT SYSTEM.
- SUBMITTED TO → PUSHPA MA'AM.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Book_No
Reports
Issue/Return
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Manages
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Publisher
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Name
1
Name
Staff
Staff_id
1
M
Publishes
Maintain
Keeps track of
N
N
ReserveDate
USER_ID
N
Price
Firstname
Email
Readers
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Reserve/Return Date
N
Books
Category
Name
Edition
Lastname
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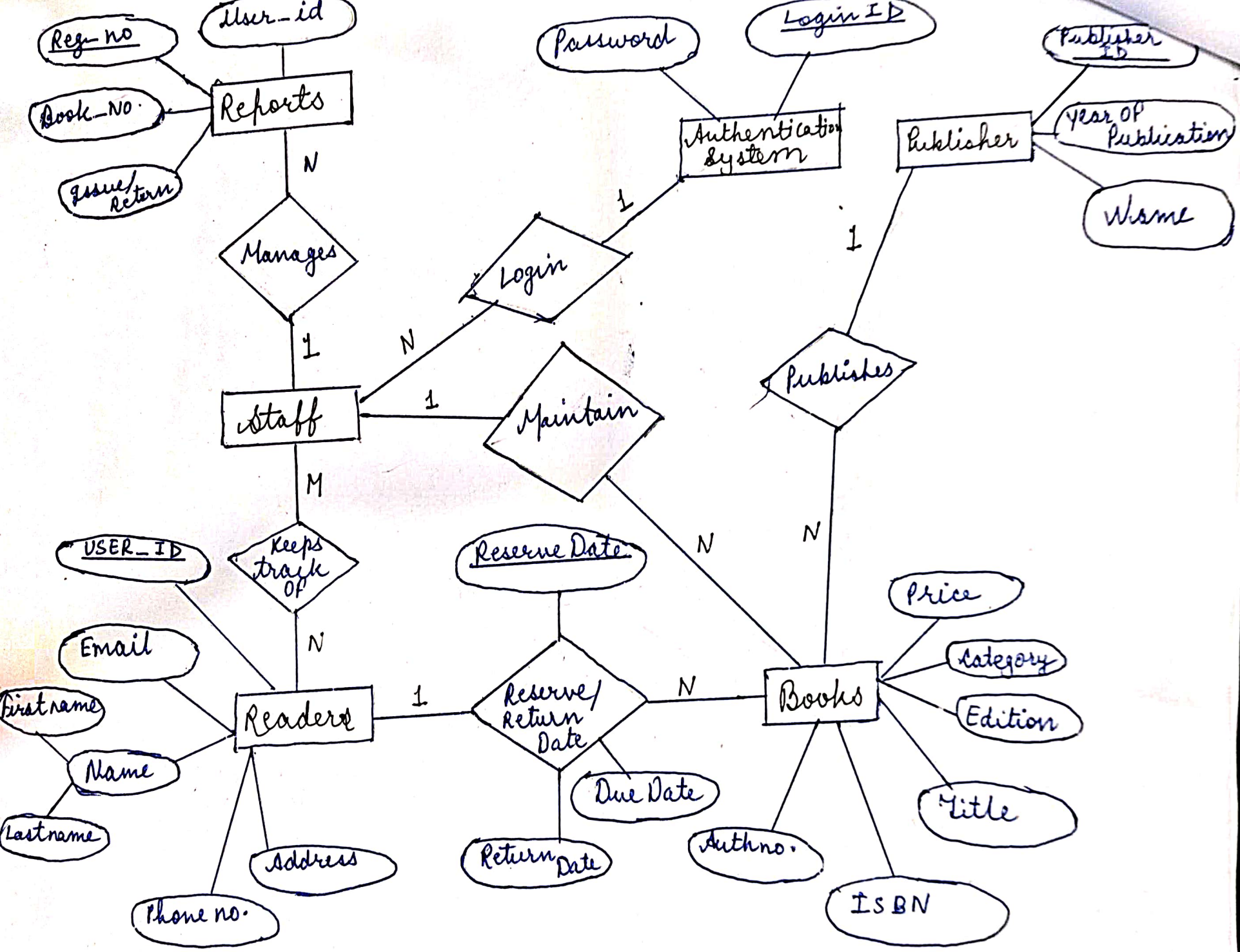
Address
Return date
Due date
Title
AuthNo
ISBN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Issue/Return
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Manages
1
Password
Login ID
Authentication System
1
Login
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Staff
1
Maintain
M
Publisher ID
Publisher
Year of Publication
Name
1
Publishes
N
N
USER_ID
Keeps track of
N
Reserve Date
Email
First name
Name
Lastname
Readers
1
Reserve/Return Date
N
Books
Price
Category
Edition
Title
Due Date
Return Date
Address
Phone no.
Authno.
ISBN

Relational schema

Main Entities → Tables

Staff
- Staff_id (PK)
- Name

Readers
- User_ID (PK)
- Firstname
- Lastname
- Name
- Email
- Phone_no
- Address

Books
- ISBN (PK)
- Title
- AuthNo
- Edition
- Category
- Price

Publisher
- Publisher_id (PK)
- Name
- Year of Publication

Authentication_System
- Login ID (PK)
- Password

Reports

- Reg-no (PK)
- User-id (FK)
- Book-No (FK)
- Issue / Return

2. Relationships → Tables / Foreign keys

1. Manages (Staff - Reports)

- one staff manages many report (1:N)

→ Add foreign key

- Report table gets Staff-id (FK)

2. Keeps Track of (Staff - Readers)

- Many-to-Many (M:N)

KeepsTrack of (
Staff-id (FK),
User-ID (FK),
PRIMARY KEY (Staff-id, User-ID)
)

3. Login (Staff - Authentication System)

- Many Staff → one login system (N:1)

Add:-

- Staff table gets LoginID (FK)

4. Maintain (Staff - Books)

- one staff maintains many books (1:N)

Add:-

- Books tables gets staff-id (FK).

5. Published (Publisher - Books)

- one Publisher → many books (1:N)

Add:

- Books table get Publisher_id (FK)

6. Reserve / Return Date (Readers - Books)

- Many-to-Many with attributes:
  - ReserveDate
  - Due date
  - Return date

Table:-

```
ReserveReturn (
    User_ID (FK),
    ISBN (FK),
    ReserveDate,
    DueDate,
    ReturnDate,
    PRIMARY KEY (User_ID, ISBN, ReserveDate)
)
```

→ Final Relational Schema

Staff (staff_id PK, Name, LoginID FK)

Readers (user_ID PK, firstname, lastname, Name, Email, Phone_no, Address)

Books (ISBN PK, Title, AuthNo, Edition, Category, Price, staff_id FK, Publisher_id FK)

Publisher (Publisher_id PK, Name, Year of Publication)

Authentication_System (LoginID PK, Password)

Reports (Reg_no PK, User_id FK, Book_No FK, Issue/Return
Staff_id FK)

KeepsTrackOf (Staff_id FK, User_ID FK, PK(Staff_id, User_ID))

ReserveReturn (User_ID FK, ISBN FK, ReserveDate, DueDate,
ReturnDate, PK(User_ID, ISBN, ReserveDate))

* Key Points to understand

- Entities → Tables
- Attributes → Colouns
- Primary key (PK) uniquely identifies records
- Foreign key (FK) Connects tables
- M:N relationships → new tables
- 1:N relationships → FK on "Many" side